

Experiment 18: Perceptron Based Iris Classification

Aim

To implement **Iris flower classification** using the **Perceptron algorithm**.

Python Program

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import Perceptron
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Load Iris dataset
iris = load_iris()

X = iris.data
y = iris.target

# Split dataset into training and testing
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Create Perceptron model
model = Perceptron(max_iter=1000, eta0=0.1, random_state=42)

# Train the model
model.fit(X_train, y_train)

# Predict test data
y_pred = model.predict(X_test)

print("Predicted Values:")
print(y_pred)
```

